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Section:2

Subject:OS

WEEK-6

Programs:

1.

```
knsat@SRITHIKA: ~  
GNU nano 8.7.1 virtual_address.c  
#include <stdio.h>  
  
int main() {  
    int x = 10;  
  
    printf("Value of x = %d\n", x);  
    printf("Virtual address of x = %p\n", (void *)&x);  
  
    return 0;  
}  
  
knsat@SRITHIKA:~$ nano virtual_address.c  
knsat@SRITHIKA:~$ gcc virtual_address.c -o virtual_address  
knsat@SRITHIKA:~$ ./virtual_address  
Value of x = 10  
Virtual address of x = 0x7ffdf01a7d64
```

2.

```
knsat@SRITHIKA: ~  
GNU nano 8.7.1 memory_segments.c  
#include <stdio.h>  
#include <stdlib.h>  
  
int global = 10;           // Initialized Data segment  
int bss_variable;          // Uninitialized Data (BSS) segment  
  
int main() {  
    int local = 20;         // Stack segment  
  
    int *heap = malloc(sizeof(int)); // Heap segment allocation  
    if (heap == NULL) {  
        return 1;  
    }  
    *heap = 30;  
  
    printf("Code address : %p\n", (void *)main);  
    printf("Data address : %p\n", (void *)&global);  
    printf("BSS address : %p\n", (void *)&bss_variable);  
    printf("Heap address : %p\n", (void *)heap);  
    printf("Stack address : %p\n", (void *)&local);  
  
    free(heap);  
    return 0;  
}
```

```

knsat@SRITHIKA:~$ nano memory_segments.c
knsat@SRITHIKA:~$ gcc memory_segments.c -o memory_segments
knsat@SRITHIKA:~$ ./memory_segments
Code address   : 0x5751dc3cf1a9
Data address   : 0x5751dc3d2010
BSS address    : 0x5751dc3d2018
Heap address   : 0x5751fcbf9010
Stack address  : 0x7fffc399539c

```

3.

```

GNU nano 8.7.1 page_table_simulation.c
#include <stdio.h>

int main() {
    int page_table[4] = {1, -1, 3, -1};
    int page;

    printf("Enter page number (0-3): ");
    scanf("%d", &page);

    if (page_table[page] == -1) {
        printf("Page Fault!\n");
        printf("Page must be loaded into memory.\n");
    }
    else {
        printf("Page is already in memory.\n");
        printf("Physical frame = %d\n", page_table[page]);
    }

    return 0;
}

knsat@SRITHIKA:~$ nano page_table_simulation.c
knsat@SRITHIKA:~$ gcc page_table_simulation.c -o page_table_simulation
knsat@SRITHIKA:~$ ./page_table_simulation
Enter page number (0-3): 0
Page is already in memory.
Physical frame = 1

```

4.

```
knsat@SRITHIKA: ~  
GNU nano 8.7.1 address_translation.c  
#include <stdio.h>  
  
int main() {  
    int page_table[] = {3, 5, 1, 7};  
    int page_size = 100;  
  
    int virtual_address = 250;  
  
    int page = virtual_address / page_size;  
    int offset = virtual_address % page_size;  
  
    int frame = page_table[page];  
  
    int physical_address = frame * page_size + offset;  
  
    printf("Virtual Address = %d\n", virtual_address);  
    printf("Page Number = %d\n", page);  
    printf("Offset = %d\n", offset);  
    printf("Frame Number = %d\n", frame);  
    printf("Physical Address = %d\n", physical_address);  
  
    return 0;  
}
```

```
knsat@SRITHIKA:~$ nano address_translation.c  
nano address_translation.c  
  
knsat@SRITHIKA:~$ gcc address_translation.c -o address_translation  
knsat@SRITHIKA:~$ ./address_translation  
Virtual Address = 250  
Page Number = 2  
Offset = 50  
Frame Number = 1  
Physical Address = 150
```

5.

```
knsat@SRITHIKA: ~  
GNU nano 8.7.1 address_translation.c  
#include <stdio.h>  
  
int main() {  
    int page_table[] = {3, 5, 1, 7};  
    int page_size = 100;  
  
    int virtual_address = 250;  
  
    int page = virtual_address / page_size;  
    int offset = virtual_address % page_size;  
  
    int frame = page_table[page];  
  
    int physical_address = frame * page_size + offset;  
  
    printf("Virtual Address = %d\n", virtual_address);  
    printf("Page Number = %d\n", page);  
    printf("Offset = %d\n", offset);  
    printf("Frame Number = %d\n", frame);  
    printf("Physical Address = %d\n", physical_address);  
  
    return 0;  
}
```

```
knsat@SRITHIKA:~$ nano address_translation.c  
knsat@SRITHIKA:~$ nano dynamic_array.c  
knsat@SRITHIKA:~$ ^C  
knsat@SRITHIKA:~$ gcc dynamic_array.c -o dynamic_array  
knsat@SRITHIKA:~$ ./dynamic_array  
10 20 30 40 50
```